DEMO DATA

Demo Datasets

Exactly which public data the MIRQAB evaluator demo uses, where it comes from, how it enters the platform, and what MIRQAB actually derives from it.

TEAM

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PROJECT

MIRQAB — AI-Powered Security Operations Platform for SMBs

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Principles

- The evaluator demo imports **only public, license-cleared, open-source sample data** that is bundled in the repository. Nothing is downloaded during setup.
- **No private, company or customer data** is part of the demo dataset.
- Nothing is hand-authored in the database. Every incident, indicator and ticket is whatever MIRQAB's own pipeline derives from the dataset rows, and the response-action history starts empty.
- Imported incidents are labelled **Open-Source Dataset** with their origin, so they are never presented as live telemetry. DEMO

Datasets used by the evaluator demo

1. OTRF Security-Datasets — APT29 emulation plan

Dataset name	APT29 emulation plan (MITRE ATT&CK Evaluations, Round 2), condensed attack chain
Source	`https://github.com/OTRF/Security-Datasets` — `datasets/compound/apt29/emulationplans/apt29.xlsx`. Pinned commit of the license file: `d9d40ef123d2c87d5d3df28c96bcab4f0faccc87`
Files in MIRQAB	`datasets/Security-Datasets/apt29-attack-chain.json` (7.2 KB, 22 steps) and the original `apt29-emulation-plan.xlsx` (25 KB)
Purpose	A realistic, multi-host intrusion story (initial breach → collection and exfiltration → persistence → credential access → lateral movement) that exercises detection, correlation and the attack-path view
Format	JSON: `{step, stage, techniques[], source, target, description}`. 22 steps, 13 stages, hosts attacker → SCRANTON, NASHUA, UTICA, NEWYORK. Condensed by hand from the spreadsheet: step IDs, stages, technique IDs and hosts are kept as-is, while verbose command text and credentials were dropped.
How it enters MIRQAB	`DemoDataService` (enabled by `MIRQAB_DEMO_DATA=true`) sends each step through the normal analysis pipeline (`AiService.analyzeLog`) with source `OPEN_SOURCE_DATASET`. The same data also drives the static **Attack Path** view in The Citadel.
What MIRQAB extracts	Severity, MITRE technique (where one of the 12 curated techniques matches), IOC/IOA and persistence findings, correlation into cases, auto-tickets for HIGH/CRITICAL, notifications
MITRE relevance	The plan references 40 distinct ATT&CK technique IDs (some are older IDs since renamed in later ATT&CK versions). MIRQAB's rule set covers 12 techniques, so only part of the chain maps to a technique; the rest stays `UNCLASSIFIED`.
License / provenance	**MIT** — `datasets/Security-Datasets/LICENSE` ("Copyright (c) 2021 Open Threat Research Forge"). The dataset notes record that the repository page's GPL badge is stale and that the root `LICENSE` file is MIT.

Note: the dataset notes (`NOTES.md`) say "21 of the 30+ steps", but the file actually contains 22 steps, which matches the demo README and the import result below.

2. PcapMonkey — Zeek sample events

Dataset name	PcapMonkey Zeek expected-output sample (6 records)
Source	`https://github.com/certego/PcapMonkey` — its test fixture `test/expected_zeek.json` (615 records upstream). Pinned commit of the license file: `5e631f203ae6101a968b1607f2d7eed28a4602b4`
File in MIRQAB	`datasets/PcapMonkey/zeek-sample-events.json` (35 KB)
Purpose	Network-flow telemetry, one record per Zeek log type: `zeek.connection`, `zeek.dns`, `zeek.http`, `zeek.notice`, `zeek.ssl`, `zeek.weird`
Format	Elastic Common Schema (ECS) JSON: `_source.event.dataset`, `source/destination.{ip,port}`, `network.{bytes,transport}`, `zeek.<type>.*`
How it enters MIRQAB	`DemoDataService` → `MonitoringService.ingestPcap`, the same path as `POST /monitoring/pcap`. A parser maps ECS fields to a `PcapEventDto` (addresses, ports, transport, a one-line protocol summary, timestamp).
What MIRQAB extracts	Source/destination IPs as indicators, a protocol summary as evidence, and rule-based classification
MITRE relevance	Low. None of the 12 curated technique rules is network-flow specific, and the dataset notes document all 6 records landing as `UNCLASSIFIED` / LOW. This is reported honestly, not tuned away.
License / provenance	**GPL-3.0-or-later** — `datasets/PcapMonkey/LICENSE.txt` plus the full text in `COPYING`

3. MITRE ATT&CK technique reference

The 12 technique IDs, names and tactics built into the backend classifier (`mitre-reference.data.ts`) come from `https://attack.mitre.org`, used under the MITRE ATT&CK Terms of Use (attribution). This is reference data, not event data.

What the demo import actually produced

On **27 September 2026** the demo was run twice on the development machine, each time against a **fresh, empty database**:

1. **Backend run:** the demo backend configuration (`MIRQAB_DEMO_DATA=true` , endpoint control and external AI off) in throwaway containers.
2. **Full end-to-end run:** the complete `infra/demo` stack started from the packaged source zip, using the documented one command. Every service (gateway, dashboard, backend, PostgreSQL, Redis) was running, and the demo was used through the browser at `http://localhost:8080` .

Both runs logged "*Demo data ready: apt29 22/22 imported, pcap 6/6 imported*" and produced **identical** results:

Item	Count	Notes
Dataset events imported	28	22 APT29 steps + 6 Zeek records
Incidents	27	All with source `OPEN_SOURCE_DATASET` . Correlation merged related events, giving 28 evidence rows across 27 cases.
By severity	HIGH 7 · MEDIUM 9 · LOW 11	No CRITICAL
By MITRE technique	T1059 ×4 · T1567 ×3 · T1204 ×3 · T1105 ×2 · T1053 ×2 · T1068 ×2 · UNCLASSIFIED ×11	Rule coverage is limited to 12 techniques
Indicators	16	Types present: IP, DOMAIN, BEHAVIOR
Persistence findings	3	
Tickets	7	Auto-created for HIGH/CRITICAL incidents (7 HIGH incidents were produced)
Response actions	0	None fabricated. The demo cannot isolate endpoints.
Devices / departments	20 / 7	15 seeded virtual devices + 5 dataset hosts; 5 standard + 2 demo departments
Users	1 or 3	The demo admin, plus the optional analyst and technician when their credentials are set (as in the full run)

What the full end-to-end run confirmed

Check	Result
RBAC with the demo accounts	The IT Technician saw only the APT29 demo department (22 of 27 incidents) and received 403 for an incident in another department. The Security Analyst saw all 27.

Check	Result
Investigation data	Incident history, investigation, indicators, persistence findings, agent decisions and similar incidents all returned successfully
Reports	Case report PDF (6 pages, all 11 sections) and weekly SOC report PDF (15 pages) generated from the demo data
Response window	The 5 unacknowledged HIGH incidents were recorded as "Response window expired" and then "Automated containment unavailable: no endpoint-control provider configured", with 5 "containment result: UNAVAILABLE" notifications. The acknowledged incident was recorded as responded within the window.
Isolation safety	A manual isolation request stayed PENDING, and **0 devices** were ever marked isolated

Scope of this verification

Both runs used the development machine, where Docker and the base images were already available. The demo has **not** been run on a separate clean machine. Counts may differ if the detection rules change.

Demo hosts

Hostnames and the PcapMonkey IP (10.12.1.101) come from the datasets. The APT29 plan names hosts but not their addresses or operating systems, so SCRANTON, NASHUA, UTICA and NEWYORK are given RFC 5737 documentation addresses (192.0.2.11–14, never routable) and the OS "Not provided by dataset". These devices have no agent and cannot be isolated.

Other datasets in the repository (not used by the demo)

Folder	Source	License	Status in this package
`wazuh-sample-logs/`	geercaceres/wazuh-sample-logs	None found	**Excluded** from the source package: not redistributable
`loghub/`	logpai/loghub	Research / academic use only	**Excluded** from the source package
`EVTX-to-MITRE-Attack/`	mdecrevoisier/EVTX-to-MITRE-Attack	CC0-1.0	Included. Not used by the demo (decoding `.evtx` needs Windows tooling that is not in the Linux container).
`attack_data/`	splunk/attack_data	Apache-2.0	Included (notes only): taxonomy reference, no telemetry
`botsv3/`	splunk/botsv3	CC0-1.0	Included (notes + a clearly-marked mock chain). Not used by the demo.

Simulation Mode data is generated at runtime by the dashboard (5 synthetic scenarios using RFC 5737 documentation IPs) and is labelled "Simulation (Synthetic)". It is not a dataset.